

Measurement of the difference in Raman scattering intensity for right- and left-circularly polarized light as a function of Raman shift gives the branch of spectroscopy called *Raman optical activity* (ROA). ROA gives information about molecular configuration and conformation.

20. 15 PHOTOCHEMISTRY

Photochemistry

Photochemistry is the study of chemical reactions produced by light. Absorption of a photon of light may raise a molecule to an excited electronic state, where it will be more likely to react than in the ground electronic state. In **photochemical reactions**, the activation energy is supplied by absorption of light. In contrast, the reactions studied in Chapter 16 are **thermal reactions**, in which the activation energy is supplied by intermolecular collisions.

The energy of a photon is $E_{\text{photon}} = h\nu = hc/\lambda$. The energy of one mole of photons is $N_A h\nu$. We find the following values of photon wavelength, energy, and molar energy:

λ/nm	200 (UV)	400 (violet)	700 (red)	1000 (IR)	
$E_{\text{photon}}/\text{eV}$	6.2	3.1	1.8	1.2	(20.78)
$N_A h\nu/(\text{kJ/mol})$	598	299	171	120	

Since it usually takes at least $1\frac{1}{2}$ or 2 eV to put a molecule into an excited electronic state, photochemical reactions are initiated by UV or visible light.

Ordinarily, the number of photons absorbed equals the number of molecules making a transition to an excited electronic state. This is the **Stark–Einstein law** of photochemistry.

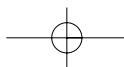
In exceptional circumstances, this law is violated. A high-power laser beam provides a very high density of photons (Prob. 20.89). There is some probability that a molecule will be hit almost simultaneously by two laser-beam photons, producing a transition in which a single molecule absorbs two photons at once. In rare cases, a single photon can excite two molecules in contact with each other. Liquid O_2 is light blue because of absorption of 630-nm (red) light; each photon absorbed excites two colliding O_2 molecules to the lowest-lying O_2 excited electronic state [E. A. Ogryzlo, *J. Chem. Educ.*, **42**, 647 (1965)].

Photochemical reactions are of tremendous biological importance. Most plant and animal life on earth depends on photosynthesis, a process in which green plants synthesize carbohydrates from CO_2 and water:



The reverse of this reaction provides energy for plants and animals. For (20.79), ΔG° is 688 kcal/mol, so the equilibrium lies far to the left in the absence of light. The presence of light and of the green pigment chlorophyll makes reaction (20.79) possible. Chlorophyll contains a conjugated ring system that allows it to absorb visible radiation. The main absorption peaks of chlorophyll are at 450 nm (blue) and 650 nm (red). Photosynthesis requires eight photons per molecule of CO_2 consumed.

The process of vision depends on photochemical reactions, such as the dissociation of the retinal pigment rhodopsin after it absorbs visible light. Other important photochemical reactions are the formation of ozone from O_2 in the earth's stratosphere, the formation of photochemical smog from automobile exhausts, the reactions in film photography, and the formation of vitamin D and skin cancer by sunlight.



Photochemical reactions are more selective than thermal reactions. By using monochromatic light, we can excite one particular species in a mixture to a higher electronic state. (The monochromaticity, high power, and tunability of lasers make them ideal sources for photochemical studies.) In contrast, heating a sample increases the translational, rotational, and vibrational energies of all species. Organic chemists use photochemical reactions as a tool in syntheses.

Certain chemical reactions yield products in excited electronic states. Decay of these excited states may then produce emission of light, a process called **chemiluminescence**. Fireflies and many deep-sea fish show chemiluminescence. In a sense, chemiluminescence is the reverse of a photochemical reaction.

Consequences of Light Absorption

Let B^* and B_0 denote a B molecule in an excited electronic state and in the ground electronic state, respectively. The initial absorption of radiation is $B_0 + h\nu \rightarrow B^*$. In most cases, the ground electronic state is a singlet with all electron spins paired. The selection rule $\Delta S = 0$ (Sec. 20.11) then shows that the excited electronic state B^* is also a singlet.

Following light absorption, many things can happen.

The B^* molecule is usually produced in an excited vibrational level. Intermolecular collisions (especially collisions with the solvent if the reaction is in solution) can transfer this extra vibrational energy to other molecules, causing B^* to lose most of its vibrational energy and attain an equilibrium population of vibrational levels, a process called **vibrational relaxation**.

The B^* molecule can lose its electronic energy by spontaneously emitting a photon, thereby falling to a lower singlet state, which may be the ground electronic state: $B^* \rightarrow B_0 + h\nu$. Spontaneous emission of radiation by an electronic transition in which the total electronic spin doesn't change ($\Delta S = 0$) is called **fluorescence**. Fluorescence is favored in low-pressure gases, where the time between collisions is relatively long. A typical lifetime of an excited singlet electronic state is 10^{-8} s in the absence of collisions.

The B^* molecule can transfer its electronic excitation energy to another molecule during a collision, thereby returning to the ground electronic state, a process called **radiationless deactivation**: $B^* + C \rightarrow B_0 + C$, where B_0 and C on the right have extra translational, rotational, and vibrational energies.

The B^* molecule (especially after undergoing vibrational relaxation) can make a radiationless transition to a different excited electronic state: $B^* \rightarrow B^{*'}$. Conservation of energy requires that B^* and $B^{*'}$ have the same energy. Generally, the molecule $B^{*'}$ has a lower electronic energy and a higher vibrational energy than B^* .

If B^* and $B^{*'}$ are both singlet states (or both triplet states), then the radiationless process $B^* \rightarrow B^{*'}$ is called **internal conversion**. If B^* is a singlet electronic state and $B^{*'}$ is a triplet (or vice versa), then $B^* \rightarrow B^{*'}$ is called **intersystem crossing**. Recall that a triplet state has two unpaired electrons and total electronic spin quantum number $S = 1$.

Suppose $B^{*'}$ is a triplet electronic state. It can lose its electronic excitation energy and return to the ground electronic state during an intermolecular collision or by intersystem crossing to form B_0 in a high vibrational energy level. In addition, $B^{*'}$ can emit a photon and fall to the singlet ground state B_0 . Emission of radiation with $\Delta S \neq 0$ is called **phosphorescence**. Phosphorescence violates the selection rule $\Delta S = 0$ and has a very low probability of occurring. The lifetime of the lowest-lying excited triplet electronic state is typically 10^{-3} to 1 s in the absence of collisions.

[The term **luminescence** refers to any emission of light by electronically excited species, and includes fluorescence and phosphorescence (which are emissions that follow electronic excitation by absorption of light), chemiluminescence, luminescence following collisions with electrons (as in gas-discharge tubes or from television screens), etc.]

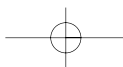


Figure 20.50

Photophysical processes. Dashed arrows indicate radiationless transitions. S_0 is the ground (singlet) electronic state. S_1 and S_2 are the lowest two excited singlet electronic states. T_1 is the lowest triplet electronic state. For simplicity, vibration–rotation levels are omitted.

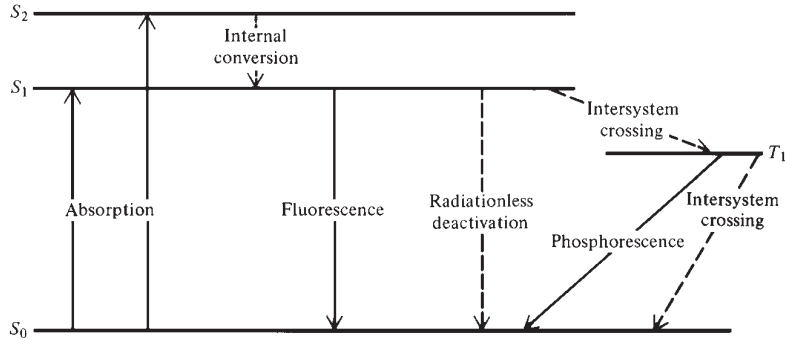


Figure 20.50 summarizes the preceding processes.

Besides the above physical processes, absorption of light can cause several kinds of chemical processes.

Since B^* is often formed in a high vibrational level, the B^* molecule may have enough vibrational energy to dissociate: $B^* \rightarrow R + S$. The decomposition products R and S may react further, especially if they are free radicals. If B^* is a diatomic molecule with vibrational energy exceeding the dissociation energy D_e of the excited electronic state, then dissociation occurs in the time it takes one molecular vibration to occur, 10^{-13} s. For a polyatomic molecule with enough vibrational energy to break a bond, dissociation may take a while to occur. There are many vibrational modes, and it requires time for vibrational energy to flow into the bond to be broken. Excitation of a diatomic molecule to a repulsive electronic state [one with no minimum in the $E_e(R)$ curve] always causes dissociation. Excitation of a diatomic molecule to a bound excited electronic state with a minimum in the $E_e(R)$ curve causes dissociation if the vibrational energy of the excited molecule exceeds D_e . See Fig. 20.51. Sometimes a molecule undergoes internal conversion from a bound excited state to a repulsive excited electronic state, which then dissociates.

The vibrationally excited B^* molecule may isomerize: $B^* \rightarrow P$. Many cis–trans isomerizations can be carried out photochemically.

The B^* molecule may collide with a C molecule, the excitation energy of B^* providing the activation energy for a bimolecular chemical reaction: $B^* + C \rightarrow R + S$.

The B^* molecule may collide with an unexcited B or C molecule to form the excimer $(BB)^*$ or the exciplex $(BC)^*$ (Sec. 20.2), species stable only in an excited electronic state. This is especially common in solutions of aromatic hydrocarbons. The excimer or exciplex may then undergo fluorescence [$(BB)^* \rightarrow 2B + h\nu$ or $(BC)^* \rightarrow h\nu + B + C]$ or nonradiative decay to $2B$ or $B + C$.

The B^* molecule may transfer its energy in a collision to another species D , which then undergoes a chemical reaction. Thus, $B^* + D \rightarrow B + D^*$, followed by $D^* + E \rightarrow$ products; alternatively, $B^* + D \rightarrow B + P + R$. This process is **photosensitization**. The species B functions as a photochemical catalyst. An example is photosynthesis, where the photosensitizer is chlorophyll.

All these chemical processes can be preceded by internal conversion or intersystem crossing, $B^* \rightarrow B^{*'}$, so that it is $B^{*'}$ that reacts.

The many possible chemical and physical processes make it hard to deduce the precise sequence of events in a photochemical reaction.

Photochemical Kinetics

A common setup for kinetic study of a photochemical reaction exposes the sample to a continuous beam of nearly monochromatic radiation. Of course, only radiation that is absorbed is effective in producing reaction. For example, exposing acetaldehyde to

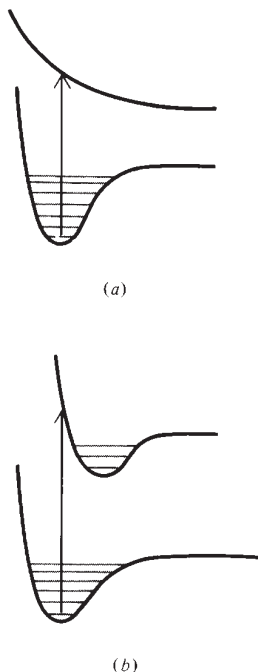
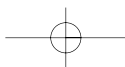


Figure 20.51

Electronic absorptions in a diatomic molecule that always lead to dissociation.



400-nm radiation will have no effect, since radiation with a wavelength less than 350 nm is required to excite acetaldehyde to a higher electronic level. According to the Beer–Lambert law (20.11), the intensity I of radiation varies over the length of the reaction cell. Convection currents (and perhaps stirring) are usually sufficient to maintain a near-uniform concentration of reactants over the cell length, despite the variation in I .

As in any kinetics experiment, one follows the concentration of a reactant or product as a function of time. In addition, one measures the rate of absorption of light energy by comparing the energies reaching radiation detectors (such as photoelectric cells) after the beam passes through two side-by-side cells, one filled with the reaction mixture and one empty (or filled with solvent only).

The initial step in a photochemical reaction is



For the elementary process (20.80), the reaction rate is $r_1 \equiv d[\text{B}^*]/dt$, where $[\text{B}^*]$ is the molar concentration of B^* . From the Stark–Einstein law, r_1 equals $\mathcal{J}_a$, where $\mathcal{J}_a$ is defined as the number of moles of photons absorbed per second and per unit volume; $r_1 = \mathcal{J}_a$. We assume that B is the only species absorbing radiation. Let the reaction cell have length l , cross-sectional area $\mathcal{A}$, and volume $V = \mathcal{A}l$. Let I_0 and I_l be the intensities of the monochromatic beam as it enters the cell and as it leaves the cell, respectively. The intensity I is the energy that falls on unit cross-sectional area per unit time, so the radiation energy incident per second on the cell is $I_0\mathcal{A}$ and the energy emerging per second is $I_l\mathcal{A}$. The energy absorbed per second in the cell is $I_0\mathcal{A} - I_l\mathcal{A}$. Dividing by the energy $N_A h\nu$ per mole of photons and by the cell volume, we get $\mathcal{J}_a$, the moles of photons absorbed per unit volume per second:

$$r_1 = \mathcal{J}_a = \frac{I_0\mathcal{A} - I_l\mathcal{A}}{VN_A h\nu} = \frac{I_0}{lN_A h\nu}(1 - e^{-\alpha[\text{B}]l}) \quad (20.81)$$

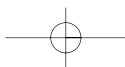
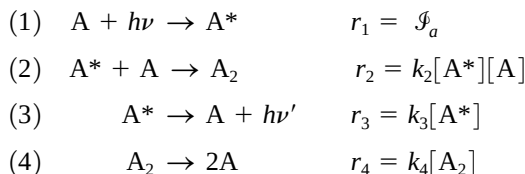
where the Beer–Lambert law (20.11) was used. In (20.81), $\alpha = 2.303\varepsilon$, where ε is the molar absorption coefficient of B at the wavelength used in the experiment.

The **quantum yield** Φ_X of a photochemical reaction is the number of moles of product X formed divided by the number of moles of photons absorbed. Division of numerator and denominator in this definition by volume and time gives

$$\Phi_X = \frac{d[\text{X}]/dt}{\mathcal{J}_a} \quad (20.82)$$

Quantum yields vary from 0 to 10^6 . Quantum yields less than 1 are due to deactivation of B^* molecules by the various physical processes discussed above and to recombination of fragments of dissociation. The quantum yield of the photochemical reaction $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$ with 400-nm radiation is typically 10^5 . Absorption of light by Cl_2 puts it into an excited electronic state that “immediately” dissociates into Cl atoms. The Cl atoms then start a chain reaction (Sec. 16.13), yielding many, many HCl molecules for each Cl atom formed.

An example of photochemical kinetics is the dimerization of anthracene ($\text{C}_{14}\text{H}_{10}$), which occurs when a solution of anthracene in benzene is irradiated with UV light. A simplified version of the accepted mechanism is:



where A is anthracene. Step (1) is absorption of a photon by anthracene to raise it to an excited electronic state; Eq. (20.81) gives $r_1 = \mathcal{J}_a$. Step (2) is dimerization. Step (3) is fluorescence. Step (4) is a unimolecular decomposition of the dimer.

The rate r for the overall reaction $2A \rightarrow A_2$ is

$$r = d[A_2]/dt = k_2[A^*][A] - k_4[A_2] \quad (20.83)$$

Use of the steady-state approximation for the reactive intermediate A^* gives

$$d[A^*]/dt = 0 = \mathcal{J}_a - k_2[A][A^*] - k_3[A^*] \quad (20.84)$$

which gives $[A^*] = \mathcal{J}_a/(k_2[A] + k_3)$. Substitution in (20.83) gives

$$r = \frac{k_2[A]\mathcal{J}_a}{k_2[A] + k_3} - k_4[A_2] \quad (20.85)$$

Note that $\mathcal{J}_a$ depends on $[A]$ in a complicated way [Eq. (20.81) with B replaced by A].

The quantum yield is given by (20.82) as

$$\Phi_{A_2} = \frac{d[A_2]/dt}{\mathcal{J}_a} = \frac{r}{\mathcal{J}_a} = \frac{k_2[A]}{k_2[A] + k_3} - \frac{k_4}{\mathcal{J}_a}[A_2] \quad (20.86)$$

If $k_4 = 0$ (no reverse reaction) and $k_3 = 0$ (no fluorescence), then Φ becomes 1. The first fraction on the right can be written as $k_2/(k_2 + k_3/[A])$; an increase in $[A]$ increases Φ , since it increases r_2 (dimerization) compared with r_3 (fluorescence). This is the observed behavior. A typical Φ for this reaction is 0.2.

Instead of dealing with the individual rates of all the physical and chemical processes that follow absorption of radiation, one often adopts the simplifying approach of writing the initial step of the reaction as



Here, R and S are the first chemically different species formed following the absorption of radiation by B. Step I really summarizes several processes, namely, absorption of radiation by B to give B^* , deactivation of B^* by collisions and fluorescence, decomposition (or isomerization) of B^* to R and S, and recombination of R and S immediately after their formation (recall the cage effect). The quantity ϕ , called the **primary quantum yield**, varies between 0 and 1. The greater the degree of collisional and fluorescent deactivation of B^* , the smaller ϕ is. For absorption by gas-phase diatomic molecules that leads to dissociation, the dissociation is so rapid that deactivation is usually negligible and $\phi \approx 1$.

The Photostationary State

When a system containing a chemical reaction in equilibrium is placed in a beam of radiation that is absorbed by one of the reactants, the rate of the forward reaction is changed, thereby throwing the system out of equilibrium. Eventually a state will be reached in which the forward and reverse rates are again equal. This state will have a composition different from that of the original equilibrium state and is a **photostationary state**. It is a steady state (Sec. 1.2) rather than an equilibrium state, because removal of the system from its surroundings (the radiation beam) will alter the system's properties. An important photostationary state is the ozone layer in the earth's stratosphere (Sec. 16.16).

20.16 GROUP THEORY

Molecular symmetry elements and operations were discussed in Sec. 20.5. The full application of molecular symmetry uses the mathematics of group theory. This section gives an introduction to group theory and omits most proofs. For fuller details, see

